Hospital Duty Roster & Shift Management System

System Overview

The duty roster system manages staff allocation across different shifts throughout a 24-hour period, ensuring adequate coverage for patient care.

1. Shift Definition & Configuration

Standard Hospital Shifts (24-hour coverage)

```
Day Shift (D):     06:00 - 14:00 (8 hours)
Evening Shift (E): 14:00 - 22:00 (8 hours)
Night Shift (N):   22:00 - 06:00 (8 hours)
```

Alternative Shift Patterns

```
12-Hour Shifts:
Day (D):    07:00 - 19:00 (12 hours)
Night (N):  19:00 - 07:00 (12 hours)

Split Shifts:
Morning (M):  06:00 - 10:00 (4 hours)
Afternoon (A): 10:00 - 14:00 (4 hours)
Evening (E):   14:00 - 18:00 (4 hours)
Night (N):     18:00 - 06:00 (12 hours)
```

2. Roster Components Analysis

From the image, I can identify these roster elements:

Staff Categories

- **Nurses** (Different grades/levels)
- **Doctors** (Various specializations)
- **Support Staff** (Cashiers, etc.)

Shift Coding System

- **D1, D2, D3** = Day shifts (different units/wards)
- **N1, N2, N3** = Night shifts (different units/wards)
- **OFF** = Day off
- **WO** = Week off
- **L** = Leave
- **H** = Holiday
- **AL** = Annual Leave

Roster Features

- **Monthly View** (July/August 2025)
- **Daily Coverage** (All 31 days)
- **Multiple Departments**
- **Staff Rotation**
- **Leave Management**
- **Overtime Tracking**

3. System Flow & Process

Phase 1: System Setup

```
1. Define Shift Types
   - Create shift templates (Day, Evening, Night)
   - Set working hours for each shift
   - Define break times and meal periods

2. Department Configuration
   - Set up departments/wards
   - Define minimum staffing requirements per shift
   - Set skill/qualification requirements

3. Staff Classification
   - Categorize staff by role (Nurse, Doctor, Support)
   - Assign skill levels and certifications
   - Set availability preferences
```

Phase 2: Roster Planning

```
1. Staffing Requirements Analysis
   - Calculate minimum staff needed per shift
   - Consider patient census and acuity
   - Account for mandatory ratios (nurse-to-patient)

2. Staff Availability Management
   - Track leave requests and approvals
   - Manage training schedules
   - Handle availability constraints

3. Roster Generation
   - Auto-assign based on rules and preferences
   - Ensure fair rotation among staff
   - Balance workload distribution
```

Phase 3: Daily Operations

```
1. Shift Handovers
   - Track who's coming on/off duty
   - Manage late arrivals and absences
   - Handle emergency replacements

2. Real-time Adjustments
   - Call-in staff for emergencies
   - Manage unexpected absences
   - Overtime authorization

3. Compliance Monitoring
```

```
   - Track working hour limits
   - Ensure rest period compliance
   - Monitor mandatory break times
```

4. Database Model Structure

Core Tables Needed

Shift Types

```
CREATE TABLE shift_types (
    id INT PRIMARY KEY,
    name VARCHAR(50),           -- "Day Shift", "Night Shift"
    code VARCHAR(10),           -- "D", "N", "E"
    start_time TIME,            -- 06:00:00
    end_time TIME,              -- 14:00:00
    duration_hours DECIMAL(4,2), -- 8.00
    is_overnight BOOLEAN,       -- TRUE for night shifts
    color_code VARCHAR(7)       -- #FF5733 for UI display
);
```

Department Shifts

```
CREATE TABLE department_shifts (
    id INT PRIMARY KEY,
    department_id INT,
    shift_type_id INT,
    min_staff_required INT,
    max_staff_allowed INT,
    skill_requirements JSON,    -- Required qualifications
    is_active BOOLEAN
);
```

Roster Periods

```
CREATE TABLE roster_periods (
    id INT PRIMARY KEY,
    name VARCHAR(100),          -- "July 2025 Roster"
    start_date DATE,            -- 2025-07-01
    end_date DATE,              -- 2025-07-31
    status ENUM('draft', 'published', 'active', 'closed'),
    created_by INT,
    published_date DATETIME
);
```

Roster Assignments

```
CREATE TABLE roster_assignments (
    id INT PRIMARY KEY,
    roster_period_id INT,
    employee_id INT,
    department_id INT,
    shift_type_id INT,
    assignment_date DATE,       -- 2025-07-15
    status ENUM('scheduled', 'confirmed', 'completed', 'absent'),
    assigned_by INT,
    notes TEXT
);
```

Leave/Absence Management

```
CREATE TABLE roster_absences (
    id INT PRIMARY KEY,
    employee_id INT,
    absence_type ENUM('leave', 'sick', 'training', 'holiday'),
    start_date DATE,
    end_date DATE,
    approved_by INT,
    status ENUM('pending', 'approved', 'denied')
);
```

5. Roster Generation Algorithm

Step 1: Gather Requirements

```
FOR each day in roster_period:
    FOR each department:
        FOR each shift:
            - Get minimum staff required
            - Get available staff pool
            - Check skill requirements
            - Consider leave/absences
```

Step 2: Fair Distribution

```
- Track consecutive shifts per employee
- Ensure weekend rotation fairness
- Balance day/night shift distribution
- Respect maximum working hours per week
- Honor leave requests
```

Step 3: Optimization Rules

```
1. Priority Rules:
   - Senior staff get first preference
   - Fair rotation among all staff
   - Minimize consecutive night shifts
   - Ensure adequate rest periods

2. Constraint Rules:
   - Maximum 5 consecutive working days
   - Minimum 11 hours between shifts
   - Maximum 48 hours per week
   - No more than 3 consecutive nights
```

6. Real-World Roster Features

Automatic Conflict Detection

- Double-booked staff
- Insufficient rest periods
- Skill mismatches
- Over/under-staffing alerts

Staff Self-Service

- View personal roster
- Request shift swaps
- Apply for leave
- Update availability

Manager Tools

- Drag-and-drop roster editing
- Real-time staffing reports
- Overtime cost tracking
- Compliance dashboards

Mobile Integration

- Push notifications for roster changes
- Emergency call-in requests
- Shift confirmation/clock-in
- Real-time staff location

7. Sample Roster Patterns

Weekly Rotation Example

```
Week 1: Day-Day-Day-Day-Day-Off-Off
Week 2: Evening-Evening-Evening-Evening-Evening-Off-Off
Week 3: Night-Night-Night-Night-Night-Off-Off
Week 4: Off-Day-Day-Evening-Evening-Night-Night
```

12-Hour Rotation Example

```
Week 1: Day-Night-Off-Day-Night-Off-Off
Week 2: Day-Night-Off-Day-Night-Off-Off
Week 3: Off-Off-Day-Night-Off-Day-Night
Week 4: Off-Day-Night-Off-Day-Night-Off
```

8. Integration Points

Payroll Integration

- Shift differentials (night/weekend pay)
- Overtime calculations
- Call allowances
- Holiday pay rates

HR Integration

- Leave balance tracking
- Training requirement alerts
- Performance management

- Compliance reporting

Hospital Operations

- Patient census data
- Emergency response teams
- Surgical schedule coordination
- Department-specific requirements

This system ensures 24/7 hospital coverage while maintaining staff welfare and operational efficiency.